Audio and Signal Processing

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1. Problem

- **Given:** $X(\omega) = |X(\omega)|\, e^{(j\angle X(\omega))}$, $\log X(\omega) = \log|X(\omega)| + j\angle X(\omega)$
- **To prove:** for a minimum-phase system, $\angle X(\omega)$ is fully determined by $\log|X(\omega)|$
- **Minimum phase:** every zero and every pole of $X(z)$ is strictly inside the unit circle

2. Setup

- **Complex cepstrum:** $\hat{x}[n] = \mathrm{IDFT}\{\log X(k)\}$
- **Step 1:** minimum phase $\Longrightarrow$ $\hat{x}[n]$ is causal ($\hat{x}[n]=0$ for $n<0$)
- **Step 2:** any real, causal $h[n]$ has $\mathrm{Re}\{H(\omega)\}$ and $\mathrm{Im}\{H(\omega)\}$ related by the discrete Hilbert transform
- **Combine:** apply Step 2 to $h[n]=\hat{x}[n]$ using Step 1's causality

3. Step 1 — Minimum Phase $\Longrightarrow$ Causal Cepstrum

$$X(z) = A \cdot \Pi_k (1 - a_k z^{-1}) / \Pi_k (1 - c_k z^{-1}),\ |a_k| < 1,\ |c_k| < 1 \text{ for all } k$$

$$\log X(z) = \log A + \Sigma_k \log(1 - a_k z^{-1}) - \Sigma_k \log(1 - c_k z^{-1})$$

$$\log(1 - a z^{-1}) = -\Sigma_{\ell=1}^{\infty} (a^\ell/\ell)\, z^{-\ell} \text{ (valid for } |z| > |a|\text{, true since } |a| < 1)$$

- **Only negative powers:** every term above is $z^{-\ell}$ ($\ell \geq 1$) — no $z^{+\ell}$ term appears

$$\log X(z) = \log A + \Sigma_{n=1}^{\infty} \hat{x}[n]\, z^{-n},\ \hat{x}[n] = (\Sigma_k a_k^n - \Sigma_k c_k^n)/n \text{ for } n \geq 1$$

- **Result:** $\hat{x}[n] = 0$ for $n < 0$ — causal. (A zero/pole outside the unit circle gives the same expansion only positive powers of z instead — an anticausal contribution.)

4. Step 2 — Causal Real Sequence $\Longrightarrow$ Hilbert Pair

- **Setup:** $h[n]$ real, $h[n]=0$ for $n<0$. Even/odd split: $h_e[n]=(h[n]+h[-n])/2$, $h_o[n]=(h[n]-h[-n])/2$

$$n > 0:\ h[-n]=0 \Longrightarrow h_e[n] = h_o[n] = h[n]/2$$

$$n < 0:\ h[n]=0 \Longrightarrow h_e[n] = h[-n]/2 = -h_o[n]$$

- **Combined:** $h_o[n] = \mathrm{sgn}(n) \cdot h_e[n]$ for all $n \neq 0$
- **DTFT:** h_e even+real $\Longrightarrow$ $H_e(\omega)$ real; h_o odd+real $\Longrightarrow$ $H_o(\omega)$ purely imaginary
- **Identify:** $\mathrm{Re}\{H(\omega)\} = H_e(\omega)$, $\mathrm{Im}\{H(\omega)\}$ comes from $H_o(\omega)$
- **Result:** multiplying by $\mathrm{sgn}(n)$ in time = Hilbert transform in frequency, so $\mathrm{Im}\{H(\omega)\} = \mathrm{Hilbert}\{\mathrm{Re}\{H(\omega)\}\}$ for any real causal $h[n]$

5. Combine

$$\mathrm{Re}\{\log X(\omega)\} = \log|X(\omega)|,\ \mathrm{Im}\{\log X(\omega)\} = \angle X(\omega)$$

$$\Longrightarrow \angle X(\omega) = \mathrm{Hilbert}\{\log|X(\omega)|\}$$

- **Conclusion:** phase is the Hilbert transform of log-magnitude for minimum-phase systems — a fixed, invertible linear map, except for $\hat{x}[0]=\log A$ (overall gain, no phase info)

6. Why Minimum Phase Is Required

- **Zero/pole outside unit circle:** contributes only to $n<0$ (anticausal) part of $\hat{x}[n]$
- **General (mixed-phase) system:** cepstrum has both causal and anticausal parts — Step 2 no longer applies
- **Consequence:** multiple phase responses can share the same $|X(\omega)|$ (all-pass ambiguity); minimum phase removes this
- **Reconstruction (Ch.6 p.31 liftering):** from $\log|X(\omega)|$ alone: IDFT → real cepstrum (even); keep $n=0$, double $n>0$, zero $n<0$ → causal $\hat{x}[n]$; DFT + exp → full $X(\omega)$

7. Numerical Verification

- **Example:** $X(z) = (1-a_1 z^{-1})(1-a_2 z^{-1})$, $a_1=0.6$, $a_2=-0.4$ (minimum phase)
- **Contrast:** a_1's zero reflected to $1/a_1$ (outside unit circle); same $|X(\omega)|$ (max diff = 7.8e-16)

Quantity	Minimum phase	Non-minimum phase (same $\|X(\omega)\|$)
max $\|\hat{x}[n]\|$, $n<0$	7.7e-18	0.400
max $\|\hat{x}[n]\|$, $n\geq 0$	0.260	0.600

- **Causality check:** min-phase $n<0$ term ≈ 0 (machine precision); non-min-phase $n<0$ term is the same order as its $n\geq 0$ term — not causal
- **Phase difference:** 3.142 rad max, despite identical magnitude

Reconstruction check (from $\log\|X(\omega)\|$ alone)	Error vs. true value
Phase	4.2e-16 rad
Log-magnitude	4.4e-16

8. Conclusion

- **Minimum phase:** all zeros/poles inside unit circle $\Longrightarrow$ cepstrum causal
- **Causal + real:** $\Longrightarrow$ Re/Im of transform are a Hilbert pair (general fact, any causal real sequence)
- **Applied to log X(ω):** $\angle X(\omega) = \text{Hilbert}\{\log|X(\omega)|\}$ — phase fully determined by log-magnitude
- **Breaks without minimum phase:** zeros/poles outside the unit circle add an anticausal cepstrum part, verified numerically above